QUALITY INFORMATION DOCUMENT

For OMI_EXTREME_MHW_ARCTIC

Issue: 1.1

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
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1) INTRODUCTION

a) Product Name

OMI_EXTREME_MHW_ARCTIC_area_averaged_anomalies

b) Indicator Evaluation Method

The time series provided here represents:

- Area-averaged temperature anomaly normalized relative to the 90th percentile of the seasonally varying climatology for the full Barents Sea region for the surface and the near bottom

A marine heatwave (MHW) index for the Barents Sea is calculated as the temperature deviation from the daily climatological average (1991-2020) normalized relative to the 90th percentile, based on model reanalysis and averaged over the area 68°N - 80°N, 18°E - 55°E. The time series of the MHW index is evaluated against normalized temperature anomalies from observations in the Kola Section in the southern Barents Sea (Alekseev, 2005; Karsakov et al., 2018; González-Pola et al., 2020), which represents the Atlantic Water inflow to the Barents Sea, and the average temperature and extent of Atlantic Water (T > 3 °C) as estimated from the annual Barents Sea Ecosystem Survey.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
ARCTIC_MULTIYEAR_PHY_002_003	QUID: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-ARC-PUM-002-ALL.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-ARC-PUM-002-ALL.pdf
ARCTIC_ANALYSISFORECAST_PHY_002_001	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-002-001.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-ALL.pdf

2) SCIENTIFIC VALIDATION

a) Barents Sea Marine Heat Wave Index

The annually averaged modelled MHW index for the full Barents Sea is statistically significantly correlated with the observed annual temperature anomaly in the Kola Section both at the surface ($R = 0.91$, $p < 0.01$) and near the bottom ($R = 0.90$, $p < 0.01$). Note that the Kola Section temperature in both comparisons represents the 0-200 m depth average.

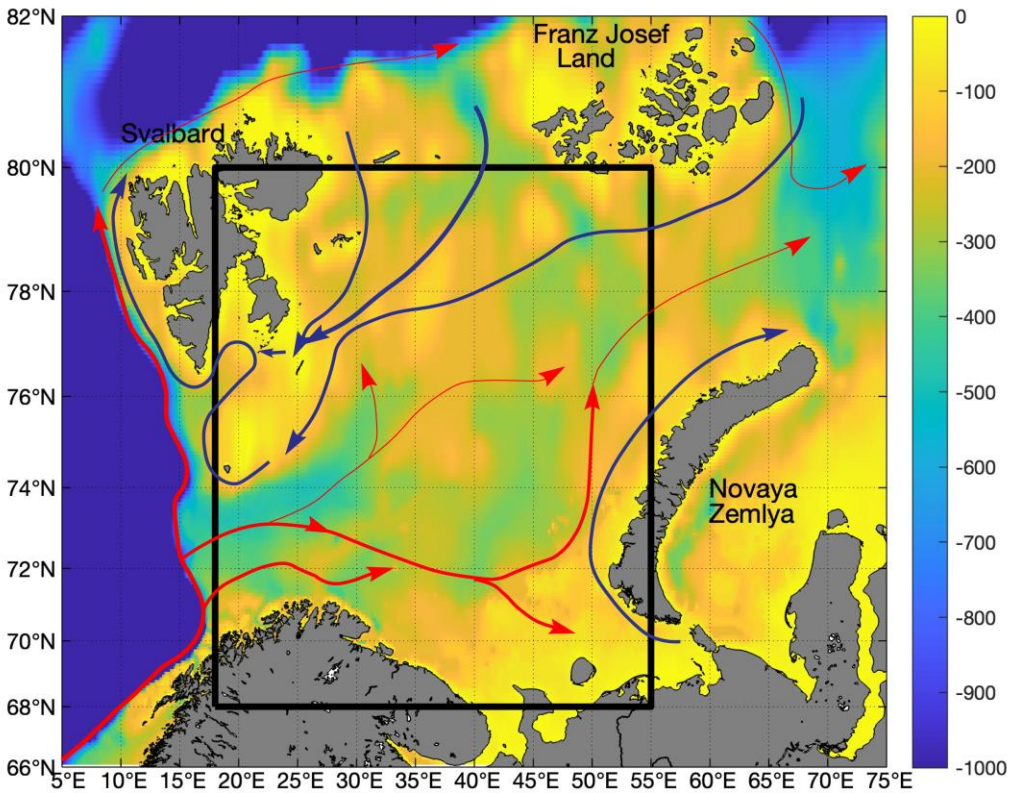


Figure 1: *Map showing the bathymetric features of the Barents Sea region. The black box shows the area used to calculate the MHW index. Red arrows denote the general circulation of warm Atlantic-derived water masses. Blue arrows denote the general circulation of cold Arctic and coastal water masses.*

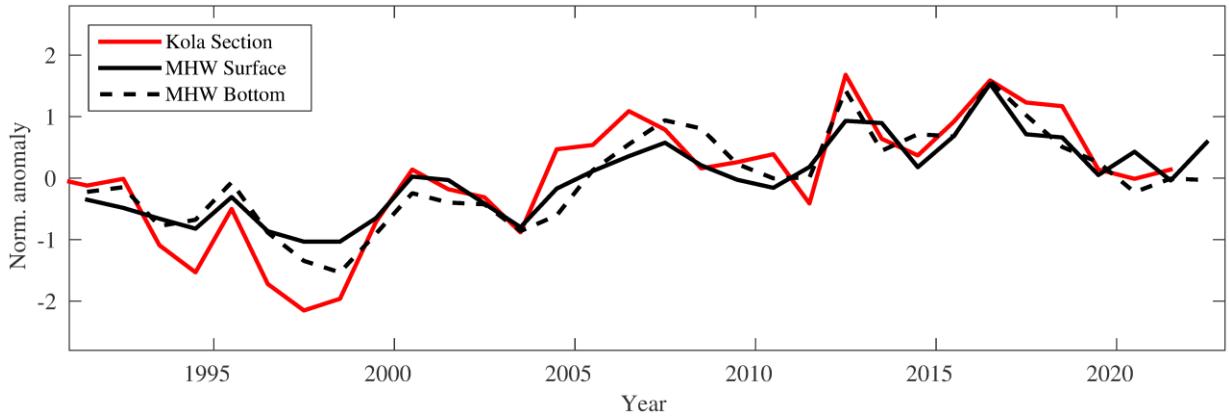
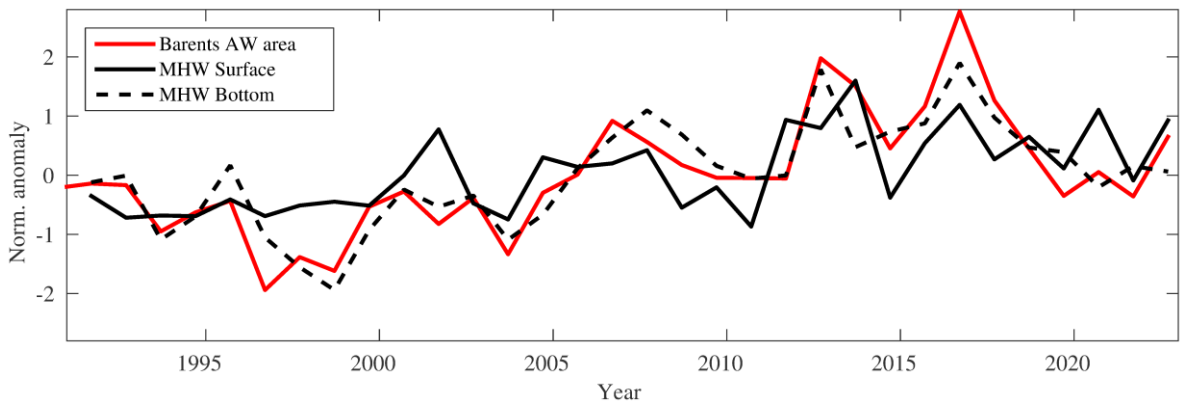


Figure 2: *Time series showing modelled MHW index in the Barents Sea for the surface (black line) and the bottom (black dashed line) and normalized observed annual temperature anomaly in the Kola Section (red line).*

For the comparison with the estimated average temperature and extent of Atlantic Water we use the September average of the MHW index because the data used for estimating the Atlantic Water temperature and extent are obtained in September every year. The observations-based estimates of Atlantic Water temperature and extent are based on Conductivity-Temperature-Depth measurements from 50-200 m depth within the area 72°N-80°N, 20°E-50°E (Johannesen et al., 2012, ICES, 2022).

The annually averaged modelled MHW index for the full Barents Sea is statistically significantly correlated with the observed Atlantic Water average temperature both at the surface ($R = 0.60$; $p < 0.01$) and near the bottom ($R = 0.93$; $p < 0.01$). The MHW index is also statistically significantly correlated with the extent of the Atlantic Water, both for the surface MHW index ($R = 0.67$; $p < 0.01$) and the bottom MHW index ($R = 0.91$; $p < 0.01$).



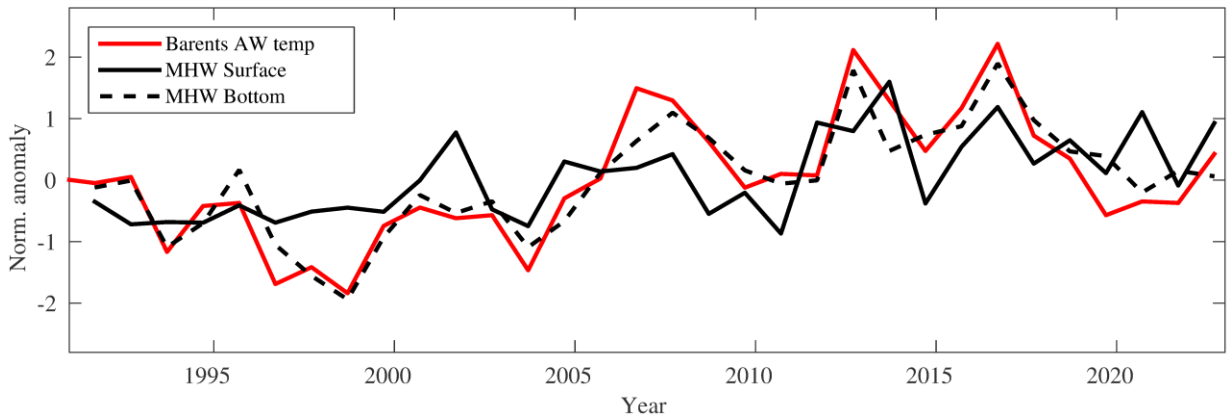


Figure 3: *Time series showing modelled MHW index in September in the Barents Sea for the surface (black line) and the bottom (black dashed line), and normalized observed average temperature (red line, top panel) and area (red line, bottom panel) of Atlantic Water in the Barents Sea in September.*

3) REFERENCES

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